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- BSc in Oceanography
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 - MSc in Oceanography
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 - PhD in Oceanography with an Interdisciplinary
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- Main interests: Deep sea, Deep-sea ecology & biology, Chemosynthetic ecosystems,
Deep-sea ecosystems connectivity, Trophic ecology, Benthos, Invertebrates, Crustaceans

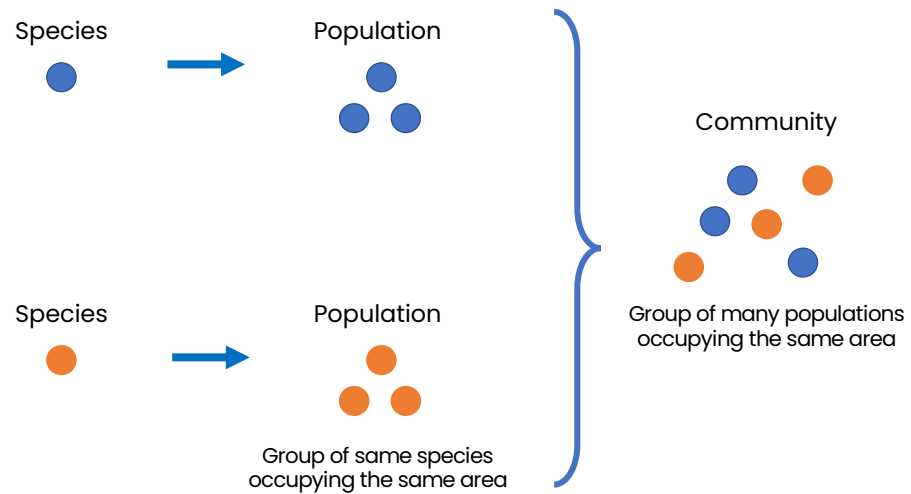
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Marine communities

- Marine community organization
- Shoreline communities
- Shallow-water communities
- Deep-sea communities

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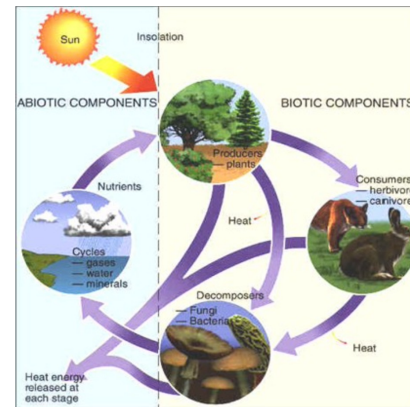
What is a community?



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Physical and biological interactions in a community

- Habitat: physical location, the living space
- Physical factors that affect the environment:
 - Temperature, depth, salinity, light exposure, etc.
- Biological factors that affect the environment:
 - Interactions between organisms (competition, predation, parasitism, commensalism, symbiosis, etc.)
- Niche: occupation within a habitat
 - Biological and physical associations
 - Use of resources, food web position
 - Function or activity



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Range of tolerance

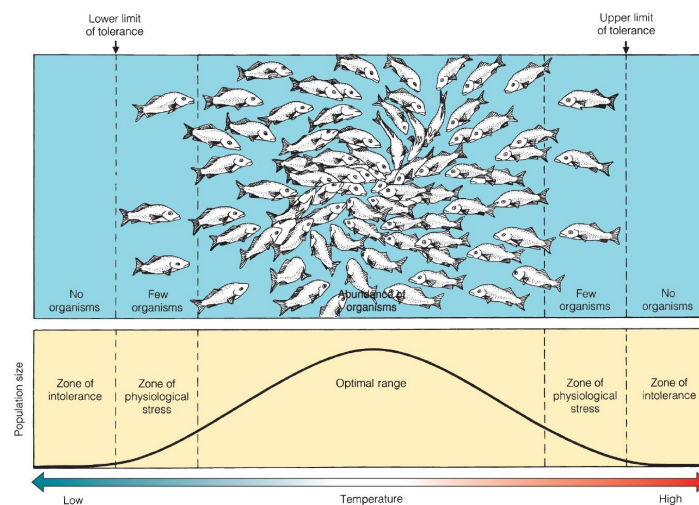
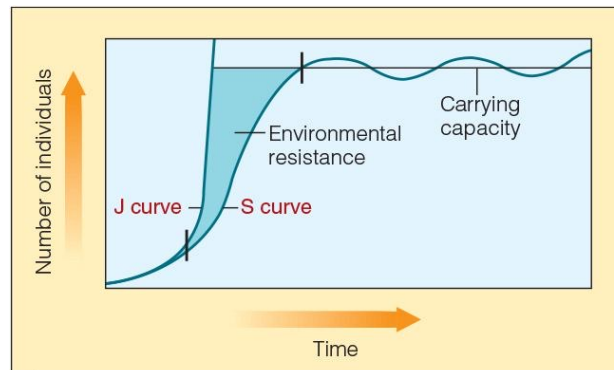


Figure 16.2 Range of tolerance to a physical factor—in this case, temperature—for a population of organisms.

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Growth rate and Carrying capacity

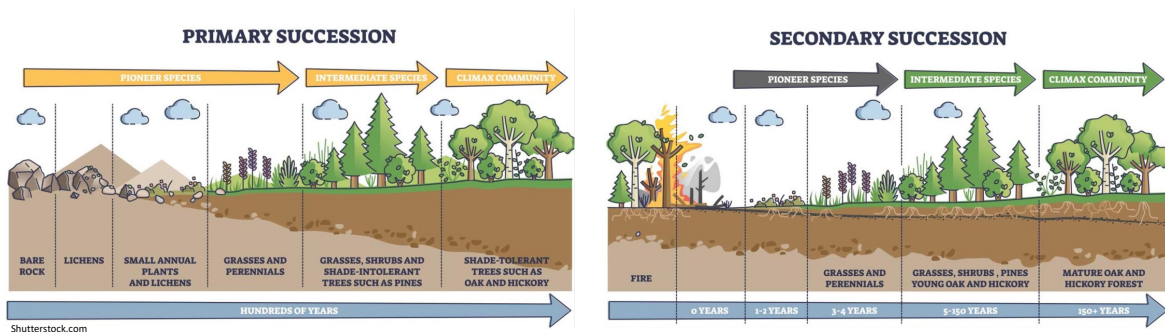
- J curve: population growth without restraints.
- S curve: population growth when there is environmental resistance.
- Carrying capacity: the number of individuals that the environment can support without degradation.



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Communities change with time

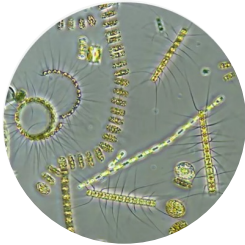
- Succession: orderly changes in time
- Climax communities: long-established, stable community



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Planktonic communities

Phytoplankton



Autotrophs (Primary Producers)
Diatoms, dinoflagellates, ciliates

Zooplankton



Heterotrophs (Consumers)

Holoplankton: spend their lives in the plankton
Copepods



Meroplankton:
temporary visitors
Mainly larvae



Photos: URI/S. Anderson, LTER-MC, R. Birke, V. Ferreira, Shutterstock.com, Microscopy UK

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Shoreline communities

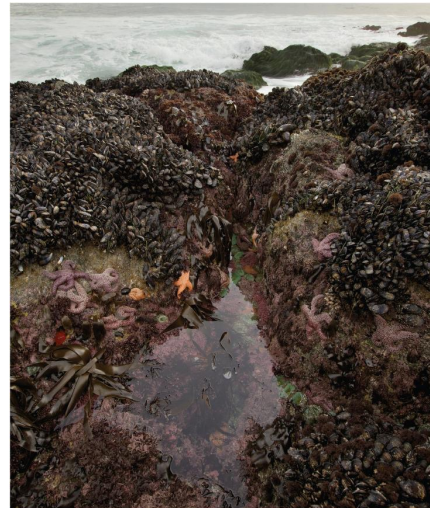
Rocky shore, Intertidal zone, Wetlands

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Physical factors affecting shoreline communities

- Wave action
- Exposure to the sun
- Substrate (sandy vs. rocky)
- Desiccation
- Salinity variation
- Temperature
- Oxygen levels

↑ or ↓ Diversity/
Density?



DAVID LUTSCHWAGER/National Geographic Image Collection

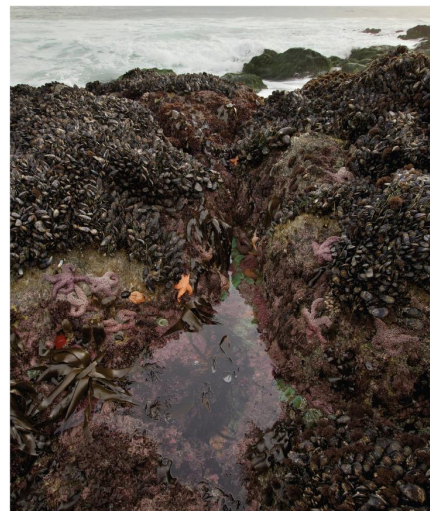
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Physical factors affecting shoreline communities

- Wave action
- Exposure to the sun
- Substrate (sandy vs. rocky)
- Desiccation
- Salinity variation
- Temperature
- Oxygen levels

↑ Diversity/Density

Highly productive,
adaptations for sun exposure
and salinity & temperature
variation

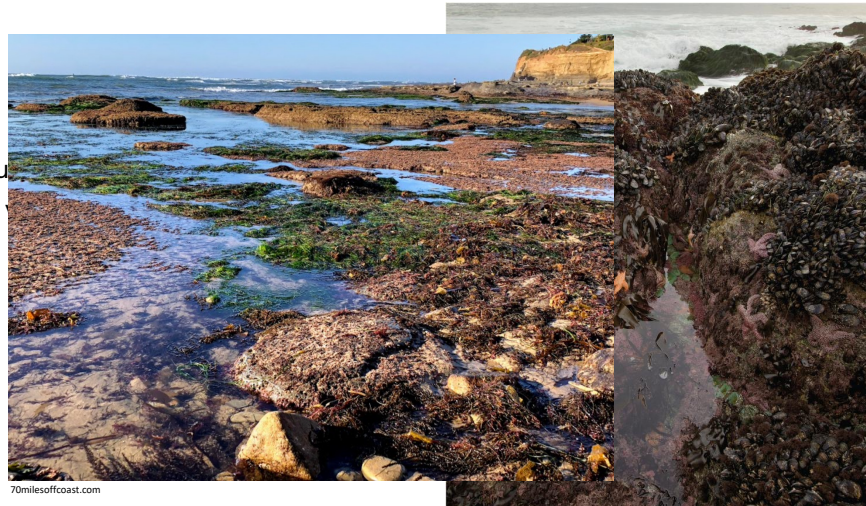


DAVID LUTSCHWAGER/National Geographic Image Collection

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Physical factors affecting shoreline communities

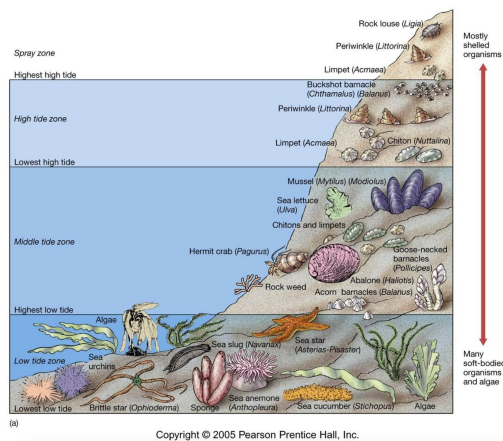
- Wave action
- Exposure to the sun
- Substrate (sandy)
- Desiccation
- Salinity variation
- Temperature
- Oxygen levels



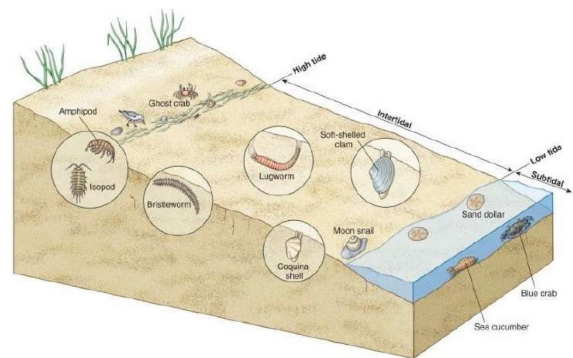
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Intertidal zonation

Rocky shores



Sandy beach



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Wetlands

Estuaries



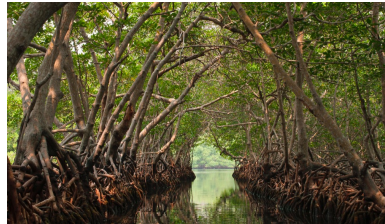
Salt marshes



Sea grasses



Mangroves



- Highly productive
- Nursery habitat
- Great diversity
- Adaptations for sun exposure and salinity variation

Photos: hhsmarine, UCSD, NOAA, Jadviga Figula

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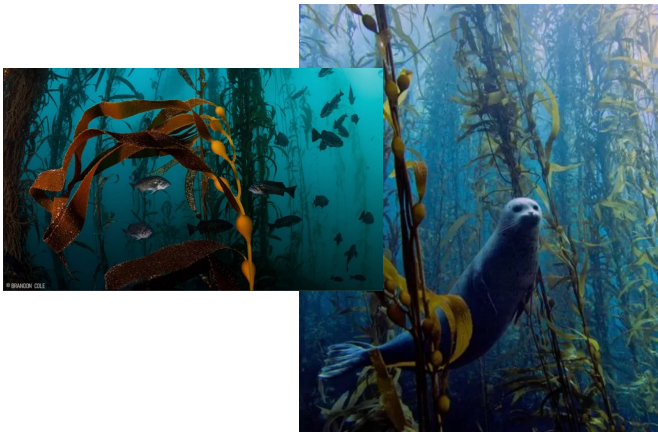
Shallow-water benthic and open-ocean communities

Kelp forests, Coral reefs, Open ocean (pelagic)

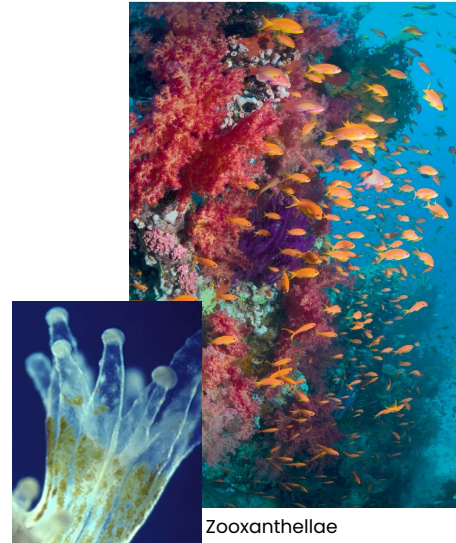
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Kelp forests and coral reefs

- Highly productive ecosystems with very dense and diverse community



Photos: B. Cole, K. McBurnie, Smithsonian

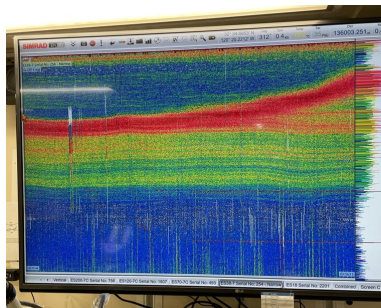


Zooxanthellae

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Open ocean (pelagic)

- 83% of total ocean biomass is concentrated in top 200 meters
 - Deep scattering layer: dense layer of animals (fish, shrimps, squids) that migrate up and down
- Diel vertical migration*



Photos: Adobe stock, Smithsonian

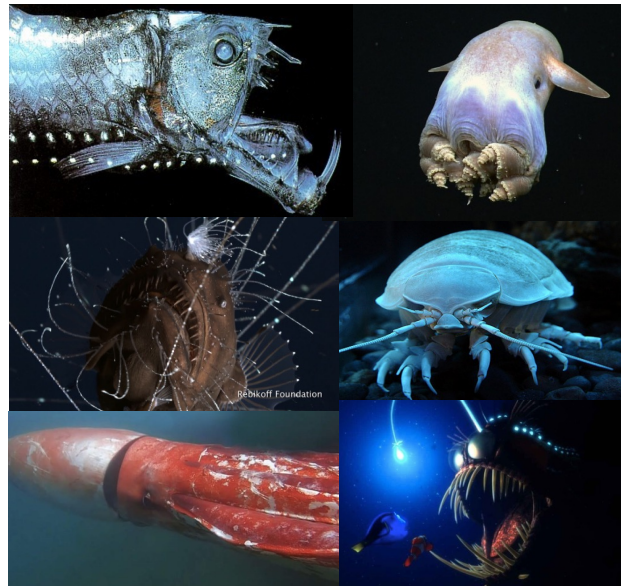
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The Deep Sea

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The deep sea

- > 200 meters deep
- Extreme temperatures (cold and hot)
- Extreme pressure
- No primary production (marine snow)
– except chemosynthesis
- No light – except bioluminescence
- Low oxygen
- Weird animals



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Adaptations to the deep sea

- No eyes or large eyes
- Bioluminescence
- Countershading
- Red color, transparent, or ultra black
- Symbiotic relationships with microbes
- Slow metabolism and growth rates
- Longevity
- **Gigantism**
- Opportunism



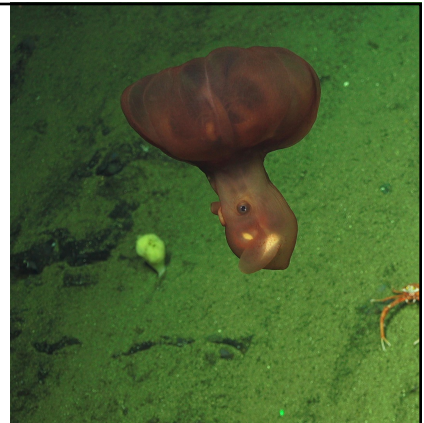
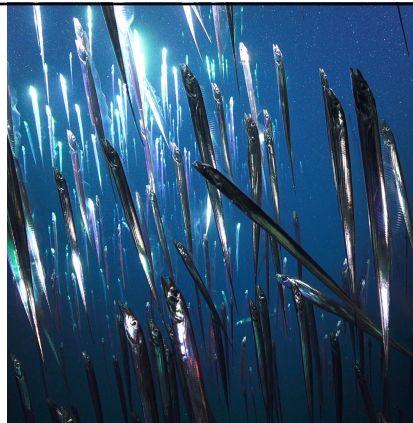
Photos: all O. S. Pereira except Pompeii worm.

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Deep-sea communities

Deep pelagic, Abyssal plain, Trenches, Seamounts and Cold-water corals, Chemosynthetic ecosystems

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Deep pelagic

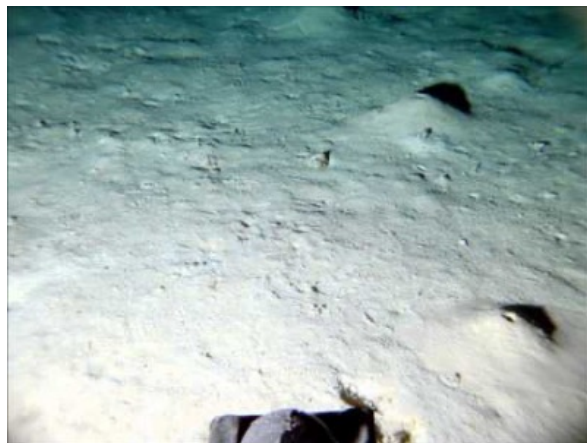
- Fairly uniform and dark
- Increasing pressure with depth
- Limited food (marine snow)
- Biodiversity is relatively low and decreases with depth
- Strategies to blend into the darkness & Bioluminescence
- Connects the deep- and shallow-water ecosystems.
- Threats: deep-sea fishing and mining.

Photos: SuBastian/Schmidt (FK190106 Levin)

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Abyssal plain

- Most of the deep ocean
- Sandy/muddy bottom
- Marine snow is the main source of food
- Not a lot of life compared to other deep-sea and shallow-water ecosystems
- Threats: deep-sea mining

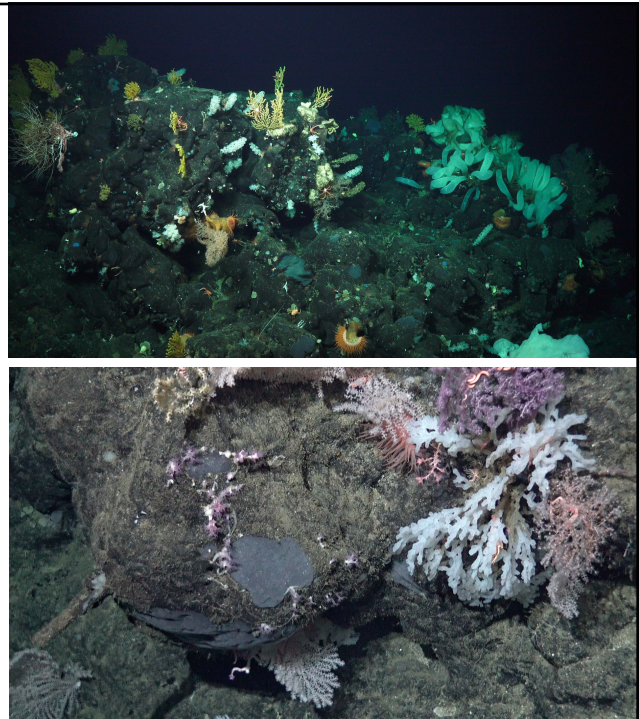


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Seamounts and Cold-water corals

- Found near boundaries of tectonic plates and mid-plate near hotspots
- Hard, rocky bottom
- High nutrient, highly productive → high biomass
- Economically important fisheries
- Cold-water corals
- Threats: deep-sea fishing, deep-sea mining, coral extraction

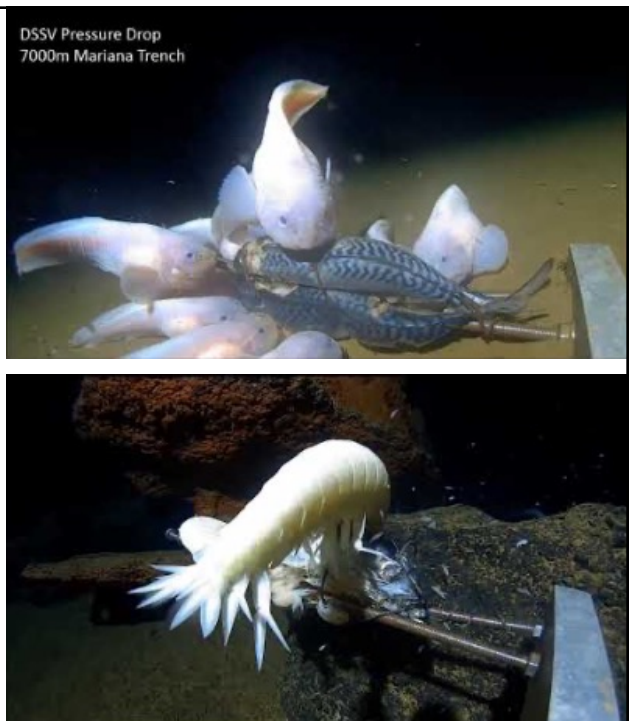
Photo & Video: SuBastian/Schmidt (FK190106 Levin)



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Trenches

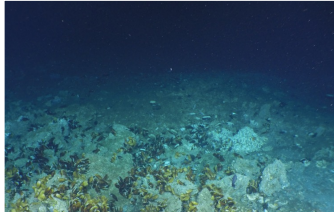
- Deepest points on Earth
- Long, narrow steep-sided depressions in the ocean
- Subduction of plates
- Pressure 1,000x greater than the surface
- Freezing temperatures
- Accumulation of food
- Deepest trench: Mariana Trench at 11,034 m
- Longest trench: Peru-Chile Trench (~5,900 km long)



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Chemosynthetic ecosystems

Methane seeps



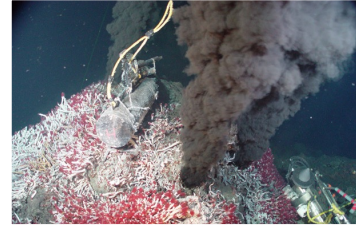
Alvin/WHOI (AT42-03/Levin)

Oxygen Minimum Zones



Gallo & Levin (2016)

Hydrothermal vents



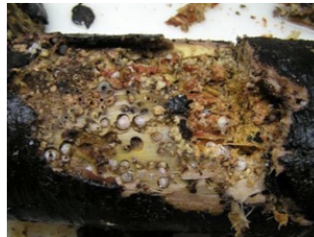
NOAA Ocean Explorer

Whale falls

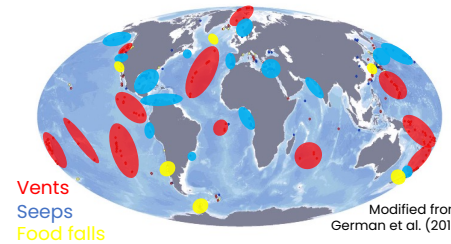


Sumida et al. (2016)

Wood falls



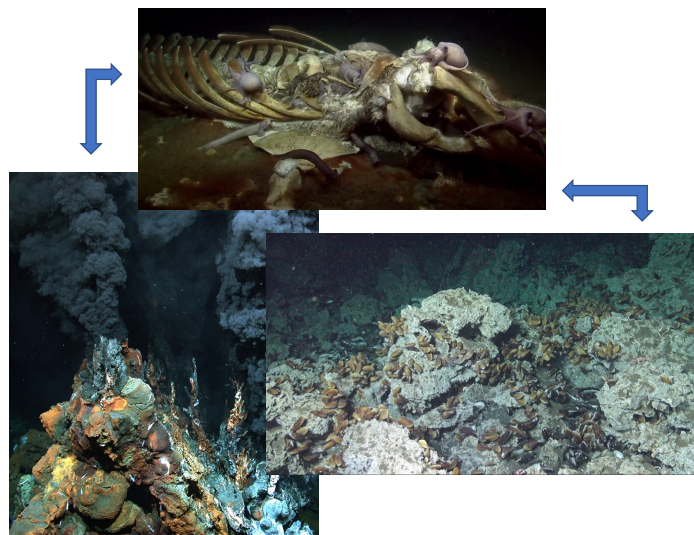
Bienhold et al. (2013)



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Connectivity among chemosynthetic ecosystems

- **Chemosynthesis:** conversion of carbon into organic matter using inorganic molecules (e.g., hydrogen, hydrogen sulfide) or methane as an energy source by microbial activity.
 - Toxic environment ($\uparrow$ sulfide)
 - Specialized, chemosynthetic communities.
 - Sporadic/temporary ecosystems.
 - Shared fauna at the genus level.
- **Stepping-stone hypothesis**



Photos: MARUM, NOAA, Alvin/WHOI (AT42-03/Alvin)

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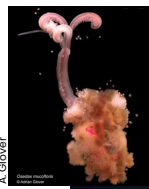
Hydrothermal vents

- Along tectonic plate boundaries (spreading and back arc basins).
- Elevated temperatures ($>400^{\circ}\text{C}$) and sulfide concentration
- Microbial mats, vesicomyid clams, giant tubeworms, Alvinocaridid shrimps, yeti crabs.
- Precipitation of minerals and rare earth elements.
- Threats: deep-sea mining.

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Organic falls

- Fresh supply of food to the deep ocean
- Other organic falls: kelp, alligator, jellyfish, turtle.



A. Glover

NOAA

Whale falls

MONTEREY BAY
NATIONAL MARINE SANCTUARY
Davidson Seamount, 3240m

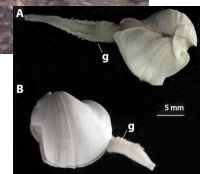
- Common along whale migration routes
- Successional stages
- Zombie worm *Osedax*
- Degradation of organic matter (both flesh and intrusion of bones) releases sulfides.

Wood falls



Alvin/WHOI (A142-03/Levin)

- Along forested margins and near the mouths of rivers
- Decay of cellulose and lignin
- Wood-boring bivalve *Xylophaga*

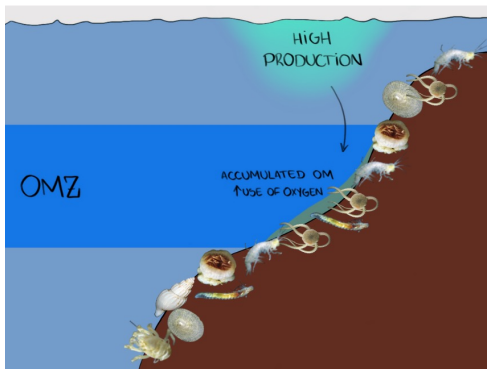


Romano et al. (2014)

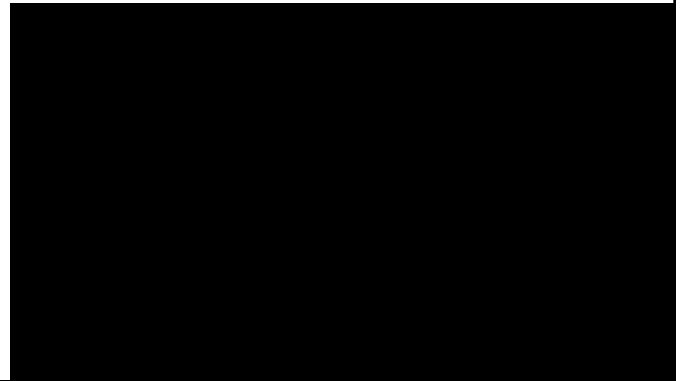
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Oxygen Minimum Zones

- On productive continental margins (upwelling)
- Usually between 400–1,000 m deep
- Decomposition of organic matter uses up oxygen
- Limited fauna (nematodes)
- Ocean deoxygenation: Expansion of OMZs



Gallo et al. (2018)



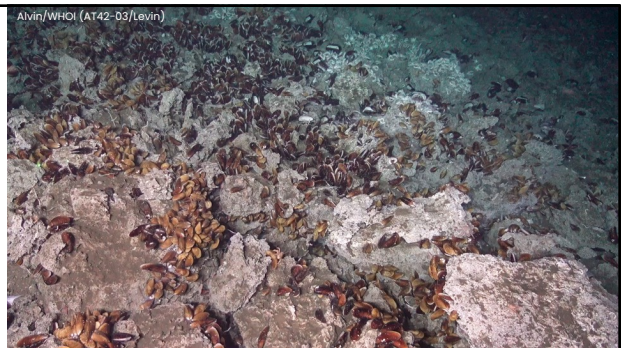
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Methane seeps

- On continental margins associated with high tectonic activity (faults).
- Energy from the oxidation of sulfide, methane and hydrogen.
- Microbial mats, bathymodiolin mussels, vesicomyid clams, siboglinid polychaetes.
- Important role in biogeochemical cycling: Climate regulation and carbon sequestration
- Threats: deep-sea fishing, oil & gas industry, ocean warming & deoxygenation



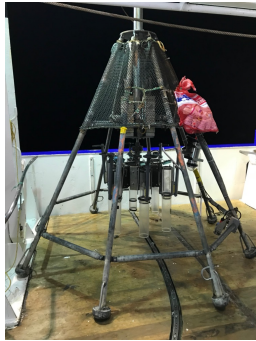
USGS



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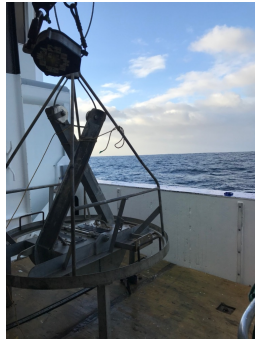
Deep-Sea Research: The basics

Multicorer



Sediment
Preserves vertical profile

Grab



Sediment
Does not preserve vertical profile

MOCNESS



Pelagic

Sediment (very destructive)
Does not preserve vertical profile

**CTD in
Rosette**

Water
Environmental data



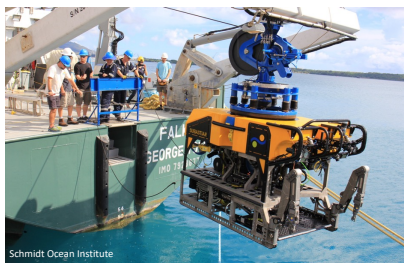
Trawl



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Deep-Sea Research: High tech

ROV SuBastian



Schmidt Ocean Institute



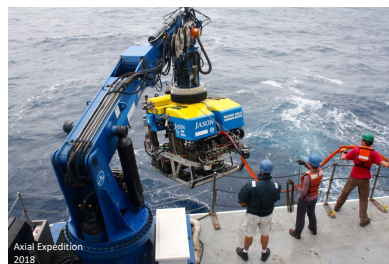
Schmidt Ocean Institute

AUV Sentry



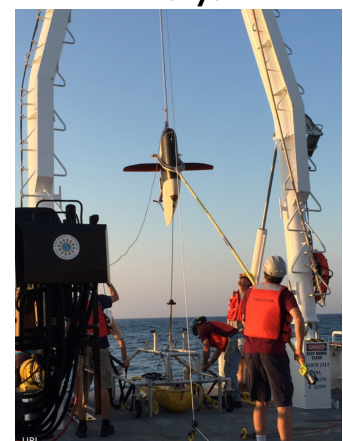
WHOI

ROV Jason



Axial Expedition
2018

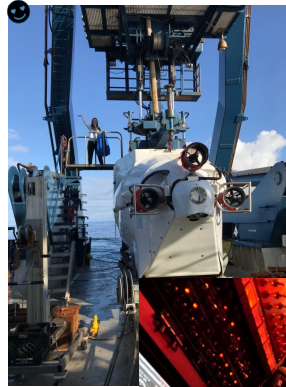
Wireflyer



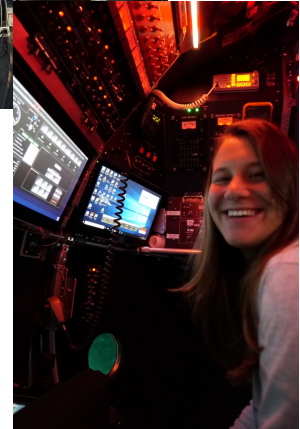
WHOI

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Deep-Sea Research: Submersibles



DSV Alvin



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Deep-Sea Research: Research ships



R/V Tangaroa, NIWA - NZ



R/V Sally Ride, SIO - EUA



R/V Atlantis, WHOI - EUA



R/V Alpha Crucis, USP - BR



R/V Roger Revelle, SIO - EUA

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Study questions

- How is a population different from a community? A niche from a habitat?
- What is a limiting factor and how is that related to range of tolerance of organisms? Give a few examples.
- How do you explain the richness and density of the intertidal zone? Relate your answer to intertidal zonation.
- What are the primary producers in kelp forests and coral reefs?
- List and explain 3 characteristics of wetlands.
- How are deep-sea ecosystems different than shallow-water ecosystems? Do they have anything in common?
- Pick 3 deep-sea ecosystems and compare them (food source, fauna, environmental conditions, etc).
- How are chemosynthetic ecosystems related? What is the stepping-stone hypothesis?
- What is deep-sea gigantism and why does it happen?
- List and explain 3 adaptations that deep-sea animals evolved mentioning what environmental factor they were overcoming.